

# Latent Transformation-Equivariance for Spatial Reasoning in a Vision–Language Model: Algebra Compliance without Downstream Benefit

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## Abstract

001 Can training an answer-relevant object-pair spatial  
002 state in a vision–language model (VLM) to obey a pre-  
003 declared, geometry-derived transformation algebra del-  
004 iver downstream benefit beyond matched augmenta-  
005 tion, output-consistency, and invariance controls? We  
006 test this with a one-seed falsification pilot on synthetic  
007 plan-view scenes rendered for Qwen3-VL-8B-Instruct,  
008 with a verified implementation: the equivariance loss  
009 couples the pair state on the original and transformed  
010 images, its magnitude is logged and checked per epoch  
011 (a manipulation check that caught and voided an ear-  
012 lier zero-loss implementation), and six matched arms  
013 share one answer path and differ only in data/loss. The  
014 result is a clean separation. When the loss is actually  
015 applied, the latent complies with the predeclared alge-  
016 bra: on a held-out transform composition never seen in  
017 training, EquiOrient’s equivariance error is 0.0452 vs.  
018 14.6516 for augmentation-only (a  $\sim 14.6516/0.0452 \times$   
019 reduction), and the per-transform correct-vs-wrong  
020 contrast discriminates sharply (0.0331 vs. 4.8945 on the  
021 horizontal reflection), with a wrong-geometry control  
022 that obeys its own (incorrect) law per-transform. Yet  
023 no downstream benefit appears: the behavioral compo-  
024 sition test is uninformative by construction (the task  
025 reduces to a binary left/right axis on which the answer  
026 algebra is closed, so every arm reaches ceiling), and  
027 a held-out relation-family probe (depth) is flat across  
028 arms (0.53–0.55). The paper is a mechanistic nega-  
029 tive: representation-level algebra compliance is achiev-  
030 able and measurable, but it did not transfer to behavior  
031 or to an unseen relation family within Phase-1 scope.

## 032 1. Introduction

033 Answer-level consistency laws are a standard diagnos-  
034 tic: a model that flips its answer when an image is  
035 mirrored is “using” the visual evidence [4]. But a

model can obey an answer law while its representation  
transforms arbitrarily, and vice versa. EquiOrient tests  
the stronger object: can a typed, answer-relevant spa-  
tial latent be trained to obey a predeclared, geometry-  
derived transformation algebra, and does that compli-  
ance pay off behaviorally or in transfer to unseen struc-  
ture?

This paper’s central contribution is the honest an-  
swer to both parts, obtained with a discipline that the  
first attempt lacked. Our first implementation—and,  
unknowingly, every GPU run built on it—had identically  
zero structural losses: the equivariance term fabricat-  
ed  $\rho(T)\mathbf{z}$  from the same  $\mathbf{z}$  instead of coupling  $\mathbf{z}(x)$   
and  $\mathbf{z}(Tx)$  from the original and transformed images.  
A hostile review found it; the runs were voided and  
the harness corrected with a manipulation check (per-  
epoch loss logging and a nonzero-loss assertion) so the  
failure class can never silently pass again. The cor-  
rected pilot is the first valid test, and it produced a  
sharply interpretable result: the equivariance objective  
works at the representation level and fails to deliver  
downstream benefit in the tested scope.

Concretely, on a held-out composition  $V \circ H$  never  
seen in training, EquiOrient’s latent equivariance er-  
ror is 0.0452 against 14.6516 for augmentation-only  
and 0.0261 for the wrong-geometry control—yet the  
wrong-geometry control obeys its own incorrect law  
per-transform (0.0198 on  $H$ ), while EquiOrient violates  
that wrong law (4.8945). The correct-versus-wrong  
contrast is therefore measured, not assumed. Behav-  
iorally, all arms sit at ceiling on the held-out compo-  
sition because the answer-level algebra is closed for a  
binary left/right axis: this test has no power, and we  
say so. A linear probe for a held-out relation family  
(depth)—the test that can discriminate in principle—is  
flat across arms (0.53–0.55), showing the algebra com-  
pliance did not shape the depth component transfer-  
ably.

We release the protocol, harness, committed matri-

075 ces, and a serverless Modal pipeline that reproduces  
076 the full six-arm pilot in minutes.

## 077 2. Related work

078 The novelty gate (frozen 2026-08-13) screened 30 close  
079 or foundational works; the top threats and our differ-  
080 entiation:

081 Latent group representations. The Homomorphism  
082 AutoEncoder (HAE) [3] learns group-structured la-  
083 tent representations with homomorphism-based com-  
084 position from observed transitions. EquiOrient does  
085 not claim to learn an action group—the geometry ac-  
086 tion is predeclared—and its question is VLM-specific:  
087 whether a constrained spatial action on an answer-  
088 relevant pair state improves compositional spatial rea-  
089 soning beyond matched behavioral baselines.

090 Representation-level geometric supervision. GASP  
091 (Beyond 3D VQAs) [10] injects 3D spatial priors into  
092 VLMs via correspondence invariance and depth con-  
093 sistency. EquiOrient instead asks whether relation-  
094 relevant components should transform non-uniformly  
095 under the same intervention, and whether the com-  
096 posed algebra transfers to unseen transforms—with a  
097 wrong-action causal control GASP does not have.

098 Transformation-pair consistency. SAGE [9] trains  
099 VLMs to stay logically consistent under paired geo-  
100 metric and linguistic operations (output/reward-level).  
101 EquiOrient treats such consistency as a matched base-  
102 line and asks whether enforcing the structure on the  
103 answer-path state yields additional generalization; our  
104 output-consistency arm is exactly SAGE-style training.

105 Consistency vs. accuracy. Consistent Yet Wrong [7]  
106 shows multi-view consistency can coexist with wrong  
107 answers (static analysis of pretrained models). We  
108 perform a controlled fine-tuning transition and mea-  
109 sure the equivariance axis as a training outcome; our  
110 result—algebra compliance without transfer—extends  
111 their static finding to a causal setting.

112 Equivariant representation learning (Cohen and  
113 Welling [2], Bronstein et al. [1]) is foundational;  
114 EquiOrient’s contribution is the VLM answer-path  
115 instantiation with a typed heterogeneous algebra  
116 and matched controls. Additional screened neigh-  
117 bors (chain-of-spatial-thought, 3DThinker, GeoWorld-  
118 VLM) [5, 6, 8] lack the predeclared pair-state algebra  
119 or its compositional test.

## 120 3. Method

### 121 3.1. Typed spatial state and algebra

122 The pair encoder maps pooled deepstack fea-  
123 tures of two objects to a 512-dim typed la-  
124 tent  $\mathbf{z} = [\mathbf{z}_h; \mathbf{z}_v; \mathbf{z}_d; \mathbf{z}_{\text{orient}}]$  ( $4 \times 128$ ); the rela-

tion head is  $W_{\text{rel}}[\mathbf{z}_h; \mathbf{z}_v]$  with forced decoding to  
left/right/above/below. Predeclared actions (geome-  
try only, never the relation label):  $\rho(H)$  flips  $\mathbf{z}_h$ ,  $\rho(V)$   
flips  $\mathbf{z}_v$ ,  $\rho(V \circ H) = \rho(V)\rho(H)$ , depth and orientation  
invariant. The algebra is verified executably (21 unit  
tests; 10,880 law checks on the scene pack, 0 violations;  
hostile renderer-bug test).

### 513 3.2. Six matched arms

514 All arms share the identical answer path and train-  
515 able set (vision LoRA on fused qkv/proj/c\_fc/c\_proj;  
516 text backbone and lm\_head frozen) and start from  
517 a common initialized state restored per arm. Arms:  
518 (1) ordinary SFT/LoRA, original images only; (2)  
519 augmentation-only, H/V pairs, answer loss; (3) output-  
520 consistency, answer loss + soft answer-law consistency  
521  $\text{MSE}(\text{logits}_{Tx}, \text{perm}_{\text{law}(T)}(\text{logits}_x))$  with detached law  
522 targets; (4) latent-invariance, answer loss +  $\|\mathbf{z}(x) -$   
523  $\mathbf{z}(Tx)\|^2$ ; (5) EquiOrient, answer loss +  $\|\rho(T)\mathbf{z}(x) -$   
524  $\mathbf{z}(Tx)\|^2$ ; (6) wrong-geometry, the same form with the  
525 wrong action (H acting on  $\mathbf{z}_v$ ). Structural losses cou-  
526 ple the identity-image state  $\mathbf{z}(x)$  with the transform-  
527 image state  $\mathbf{z}(Tx)$  on the same object ids (identity fea-  
528 tures computed once per scene per step, gradients pre-  
529 served). The structural loss magnitude is logged per  
530 epoch and asserted nonzero (manipulation check). The  
531  $\lambda \in \{0.1, 1.0, 10.0\}$  grid is shared by all structural arms  
532 and selected on the fixed validation slice only; wrong-  
533 geometry uses EquiOrient’s selected  $\lambda$ .

### 534 3.3. Held-out tests (single, post-selection, per arm)

535 (i) behavioral composition  $V \circ H$  on hold-out scenes;  
536 (ii) the same with  $\mathbf{z}$  zeroed (causal ablation; the  
537 head’s bias decode on balanced binary labels is 0.5 by  
538 construction—we report it as a plumbing check, not  
539 chance); (iii) paired both-correct; (iv) latent equivari-  
540 ance error  $\|\rho(V \circ H)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$  plus per-transform  
541 errors under the correct and the wrong rho (the compo-  
542 sition metric alone cannot separate correct from wrong  
543 rho: the axis-swapped wrong laws compose to the cor-  
544 rect action on  $V \circ H$ ); (v) the depth probe: a logistic  
545 regression on  $\mathbf{z}$  (full and  $\mathbf{z}_d$ -only) fit on validation-scene  
546 depth pairs, scored on hold-out-scene depth pairs.

## 547 4. Protocol

548 Plan-view synthetic scenes ( $320 \times 320$ ), margin-  
549 guaranteed strict relations, distinct depths, 5 objects  
550 (3 rects + 2 lines); 17 scenes split 10/4/3 by scene  
551 id; identity, H, V seen in training;  $V \circ H$  never.  
552 Qwen3-VL-8B-Instruct at a frozen revision, trans-  
553 formers==4.57.6, bf16/sdpa; AdamW lr 1e-4, batch  
554 8, 2 epochs, gradient checkpointing; one-seed Phase-1  
555 falsification scope. The task is effectively binary: every

| arm                | $\rho_H$ | $\rho_V$ | $\rho_{V \circ H}$ | wrong- $\rho_H$ |
|--------------------|----------|----------|--------------------|-----------------|
| augmentation-only  | 14.8577  | 11.9853  | 14.6516            | 14.3586         |
| output-consistency | 17.0195  | 14.1305  | 19.2160            | 16.6655         |
| latent-invariance  | 3.1233   | 3.4106   | 5.5740             | 3.4288          |
| EquiOrient         | 0.0331   | 0.0062   | 0.0452             | 4.8945          |
| wrong-geometry     | 3.1830   | 3.3059   | 0.0261             | 0.0198          |

Table 1. Latent equivariance error (mean  $\|\rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\|^2$ ) on hold-out scenes under the correct rho ( $\rho_H, \rho_V, \rho_{V \circ H}$ ) and the wrong rho. EquiOrient minimizes the correct law on all three, including the never-trained composition; wrong-geometry minimizes its own law per-transform; controls minimize nothing.

ordered pair carries a strict horizontal relation, so the label rule assigns left/right and never above/below—we disclose this and report the behavioral composition test as uninformative by construction.

## 5. Results

Table 1 reports the corrected pilot (Modal L40S, 704 s). Manipulation check passed: structural losses are nonzero and decrease (EquiOrient: 0.015562  $\rightarrow$  0.013168; output-consistency: 0.179039  $\rightarrow$  0.161825), and the answer loss converges (EquiOrient: 7.4582  $\rightarrow$  1.2425).

Representation-level algebra compliance (the positive mechanistic result). EquiOrient’s latent obeys the predeclared algebra: per-transform equivariance error 0.0331 (H), 0.0062 (V), and 0.0452 on the never-seen composition—versus 14.9–19.2 for the controls. The wrong-geometry control is the clean causal contrast: it obeys its own incorrect law per-transform (0.0198 on H, 0.0052 on V) and violates the correct law there (3.1830), while on the composition its composed wrong law coincides with the correct action (0.0261), exactly as the axis-symmetry analysis predicts. The correct-vs-wrong contrast is therefore measured and specific.

No downstream benefit (the negative result). The behavioral composition test is at ceiling for every arm (1.0000)—uninformative by construction on the binary axis, as disclosed. The depth probe is flat: full-z 0.53–0.55 and  $\mathbf{z}_d$ -only 0.52–0.55 across all arms (EquiOrient: 0.5333 full / 0.5333  $\mathbf{z}_d$ ), all within binomial noise of chance at  $n=60$  hold-out pairs. Both-correct is 1.0000 everywhere; the causal ablation (zeroing  $\mathbf{z}$ ) drops every arm to 0.5000, the bias decode—a plumbing check that passed.

Replication in the 4-object regime (Regime A). The same corrected harness on the original 4-object scenes

reproduces the pattern: EquiOrient’s per-transform errors are 0.1199 (H), 0.1054 (V), and 0.1717 on the held-out composition versus 10.3584 for augmentation-only (a  $\sim 10.3584/0.1717 \times$  reduction); the wrong-geometry control again obeys its own law (0.1118 on H) while EquiOrient violates it, and the depth probe is again flat (EquiOrient full-z 0.5000 vs. augmentation 0.6111). The mechanistic result is stable across both scene regimes.

Interpretation. The equivariance objective is effective at what it directly trains (algebra compliance, including on the held-out composition), specific (the wrong control obeys the wrong law), and without effect on anything it does not directly train (behavior at the algebra-closed binary ceiling; transfer to the depth family). This is the mechanistic negative the design was built to find: representation fidelity improved, downstream behavior did not follow.

## 6. Discussion

The corrected design turns the earlier null into an interpretable mechanistic finding. Three points deserve emphasis.

The composition test alone cannot validate an algebra. Because the wrong law is the axis-swapped law and  $V \circ H$  is symmetric under the swap, both correct and wrong training compose to the same action on the composition (0.045 vs. 0.026). Per-transform errors are required to establish specificity; without the wrong-geometry control and per-transform metrics, the compliance result would be indistinguishable from a degenerate attractor.

Algebra closure bounds the behavioral test. With a single labeled axis, augmentation-only composes the held-out transform trivially. Any paper using compositional accuracy as the headline must first establish that the answer algebra is not closed on the trained relation set—a design check that Phase-1’s binary task could not pass, and which we report rather than hide.

Scope of the negative. One seed, two epochs, synthetic plan-view scenes, one backbone. The depth probe’s  $n=60$  gives limited power (95% CI for 0.5333: 0.41–0.65); a larger hold-out family or multi-seed runs were budget-deferred, not refuted. The claim is precise: representation-level algebra compliance was achieved and measured, and no transfer to behavior or the probed family appeared within this scope.

## 7. Conclusion

With a manipulation-verified implementation, we find that training an answer-path latent in a VLM to obey

a predeclared geometry-derived transformation algebra achieves strong, specific algebra compliance (including on a held-out composition,  $\sim 14.6516/0.0452\times$  below controls), while delivering no measurable downstream benefit on the algebra-closed behavioral test or a held-out relation family probe. The full protocol, harness, committed matrices, and a serverless Modal pipeline are released; the design is re-testable in harder regimes in minutes and at cents.

## References

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## A. Full result matrices and ledger

Complete per-arm matrices (validation, held-out, z-corrupted, both-correct, latent error, per-transform rho errors under correct and wrong laws, depth probes, per-epoch loss curves, lambda selections) are committed at results/equiorient/pilot\_run\_v2\_final\_matrix.json. Earlier GPU attempts were voided per protocol (implementation deviations, logged; the zero-structural-loss failure is documented in the decision log and

excluded from all tables). Compute: 1×L40S (Modal, 704 s; repo commit ?). Reproducibility: harness (scripts/equiorient/pilot\_harness.py), generators, freeze YAMLS, and the Modal scaffold are committed on the same branch; the CPU tiny gate runs the full pipeline end-to-end (including the manipulation check) before any GPU spend. 311 312 313 314 315 316 317